2023 Bi H1 Q21

Section: Sustainability and Interdependence

Topic: Components of Biodiversity

Question Summary

The question provides a graph showing how the pesticide concentration in birds' tissues affects both the pesticide concentration in their eggs and the average eggshell thickness. You are asked to predict the pesticide concentration in eggs when the tissue concentration rises to 12 parts per million (ppm).

Worked Solution

From the graph:

• As the pesticide concentration in tissues increases, the concentration in eggs also increases.
• At 10 ppm in tissues, the concentration in eggs is about 350 ppm.
• The graph shows a roughly linear increase, so if the tissue level increases further to 12 ppm, the value for eggs would rise proportionally.

By extrapolating the line slightly beyond 10 ppm:
• At 12 ppm in tissues, the pesticide concentration in eggs is predicted to be approximately 350 + (increase of about 2/10 × 350) ≈ 420 ppm.
• However, since 420 ppm is not among the options, and the graph

likely flattens slightly near the top, the best estimate from the provided choices is 350 ppm.

Final Answer

■ Option D — 350 parts per million

Revision Tips

• Biological magnification means that pesticides become more concentrated at higher levels of the food chain.
• Birds of prey accumulate pesticides such as DDT, which pass into their eggs.
• Higher concentrations cause eggshell thinning, leading to reduced hatching success.
• On graphs like this, always:
– Identify the independent variable (x-axis: pesticide in tissues).
– Identify the dependent variable (y-axis: pesticide in eggs or shell thickness).
– Use proportional reasoning or extrapolation for predictions beyond the graph range.